

Software Engineering

Moscow Institute of Physics and Technology

01. Prelude

Introduction. Environment. Standard Template Library.

02. Functions

Types. Operators. Statements. Memory. Lvalue References. Functions.

03. Classes

Structures. Classes. Relations. Dynamic Polymorphism. Rvalue References. Operators.

04. Templates

Function and Class Templates. Forwarding References. Constant Expressions. Type Traits. Concepts.

05. Patterns

Generative, Structural and Behavioral Patterns. Static Polymorphism.

06. Translation

Preprocessing. Projects. Modules. Libraries.

07. Services

Errors. Exceptions. Debugging. Logging. Testing. Profiling.

08. Arithmetic

Bytes. Long Arithmetic. Floating - Point and Random Numbers. Time.

09. Memory

Smart Pointers. Iterators. Memory. Allocators.

10. Containers

Sequential, Nested and Associative Containers. Hash Functions. Unordered Containers.

11. Ranges

Functors. Lambda Expressions. Visitors. Ranges. Algorithms. Graphs.

12. Characters

Locales. Encodings. Strings. Regular Expressions. Parsers.

13. Streams

Streams. Iterators. Filesystem. Serialization.

14. Parallelism

Threads. Synchronization. Architectures. Atomics. Pools. Processes.

15. Networks

Networks. Transmission Control Protocol. Coroutines. Applications. Cryptography.